

Shekhar Kumar Yadav

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Professorship for Audio Information Processing
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Post-PhD Experience: I am a **postdoctoral researcher** in the Professorship for Audio Information Processing at the **Technical University of Munich (TUM)** in Germany (since July 2024). My research involves analyzing existing and developing new binaural beamforming algorithms for hearing aids that are suitable for practical dynamic acoustic environments. Previously, I was a lead researcher (**Research Associate I**) at IIT Gandhinagar on a project titled 'Development of Technique for Ground-Based Source Localization Using Sparse Array On-board LEO Satellite' funded by the Indian Space Research Organization (ISRO), Government of India, at the Indian Institute of Technology Gandhinagar from April 2023 to July 2024.

EDUCATION

Indian Institute of Technology Gandhinagar, Gandhinagar, India

PhD in Electrical Engineering

Aug. 2019 – Dec. 2023

Guide: Prof. Nithin V. George

Research Area: Array Signal Processing, Microphone Arrays, Acoustic Beamforming, Source Localization, Dereverberation and Speech Recognition

Sardar Vallabhbhai National Institute of Technology (SVNIT), Surat, India

Master of Technology in Electronics Engineering

Aug. 2017 – Jul. 2019

GPA: 9.12/10

Birla Institute of Technology and Science Pilani (BITS Pilani), Pilani Campus, Pilani, India

Bachelor of Engineering (Hons.) in Electrical and Electronics Engineering

Aug. 2011 – May. 2015

GPA: 7.71/10

Salt Brook Academy, Assam Higher Secondary Education Council, Assam, India

Higher Secondary Leaving Certificate (12th)

Aug. 2009 – Jul. 2011

Percentage: 92

Don Bosco Higher Secondary School, Board of Secondary Education, Assam, India

High School Leaving Certificate (10th)

Jan. 1997 – Mar. 2009

Percentage: 89.2

RESEARCH PROJECTS

Psychoacoustic Evaluation of Binaural Beamforming in BTE Hearing Aids in Dynamic Environment

Location: Professorship for Audio Information Processing, TU Munich

Aug. 2024 – ongoing.

- Existing binaural beamforming algorithms for BTE hearing aids were coded to run in real-time using the OpenMHA platform. Virtual audiovisual dynamic environments were created using the real-time Simulated Open Field Environment (rt-SOFE) in the anechoic chamber at TU Munich. The virtual environment ranged from being simplistic (target speaker in a circular motion and a stationary interfering speaker) to complex (an underground U-Bahn station).
- A head and torso simulator was used to study the objective performance of the binaural beamforming algorithms. Measures like ITD/ILD preservation of not only the target speaker but also of the entire acoustic scene, speech intelligibility, and signal-to-interference ratio improvement were analyzed.
- Psychoacoustic tests were performed on normal hearing participants to evaluate the real-time performance of the existing binaural beamformers.

Joint Dereverberation and Acoustic Beamforming for Robust Speech Recognition

Location: Audio Signal Processing Lab, IIT Gandhinagar

Jun. 2022 – Dec. 2022

- Developed a joint dereverberation and acoustic beamforming algorithm capable of capturing the target speech in the presence of interfering speakers and noise in a reverberant room. The proposed algorithm blindly estimates the shape parameter of a suitable prior distribution of the desired speech from the microphone data rather than keeping it fixed, resulting in a more precise speech recognition performance.
- We used maximum likelihood techniques to estimate the multi-channel linear prediction (MCLP) dereverberation coefficients and the beamforming weights. The stochastically latent precision parameters in each time-frequency (TF) bin were obtained by estimating the hyperparameters using the expectation maximization

(EM) method. For the online version of the algorithm, a recursive EM method was also proposed for real-time processing.

- The performance was evaluated not only for simulated RIRs but also for real-world RIRs from the REVERB dataset as well as for real-world recordings from the CHiME-5 challenge database.

Sparse Distortionless Modal Beamforming for Spherical Microphone Arrays

Location: Audio Signal Processing Lab, IIT Gandhinagar

Mar. 2022 – Jul. 2022

- A beamforming method with a distortionless constraint was proposed to minimise the ℓ_0 -norm of the beamformer output in the spherical harmonics (SH) domain. The proposed method assumed a super-Gaussian prior for the target speech STFT coefficients in the SH domain and sparsified the beamformer output, which was shown to reduce the residual elements of the interfering speech signals from the beamformer output. This led to superior spatial separation.

Third-Order Tensor Decomposition Based Multichannel Linear Prediction for Robust Dereverberation

Location: Audio Signal Processing Lab, IIT Gandhinagar

Apr. 2023 – Sep. 2023

- Developed a computationally efficient multichannel linear prediction based robust dereverberation method using third-order tensor decomposition of the prediction filter. One of the decomposed sub-filter was designed to act as a spatial filter, while the other two acted as temporal prediction filters. An iterative algorithm was then developed to get optimal solutions for all three sub-filters. The spatial filter was optimized as a weighted distortionless beamformer to deal with noise, while the temporal filters were optimized as weighted Wiener filters.
- Since the lengths of the three sub-filters were smaller than the actual filter, the respective covariance matrices were computationally easier to invert, leading to a computationally efficient algorithm.

Near-Field Acoustic Source Localization and Beamforming in the Spherical Sector Harmonics Domain

Location: Audio Signal Processing Lab, IIT Gandhinagar

Aug. 2023 – Dec. 2023

- This work addressed near-field acoustic localization and beamforming using a spherical sector microphone array. The representation of spherical waves from a point source in the S^2H domain using the orthonormal S^2H basis functions was presented. Using the representation, two algorithms were proposed for the joint estimation of the range, elevation and azimuth locations of near-field sources, namely NF- S^2H MUSIC and NF- S^2H MVDR. Further, a near-field beamforming algorithm capable of radial and angular filtering in the S^2H domain was also developed. Finally, the Cramer-Rao Bound (CRB) in the S^2H domain for near-field sources was presented.

Psychoacoustic Evaluation of the Effect of Active Noise Control (ANC) Headphones on the Playback Signal

Location: Audio Signal Processing Lab, IIT Gandhinagar

Aug. 2021 – Feb. 2022

- Conducted a behavioural experiment to assess if there were any differences in human listeners' perception across the 'ANC on' and 'ANC off' conditions.
- In this experiment, the participants were asked to listen to an audio instruction (e.g. "Ready Charlie go to green six now") and click on the correct rectangle out of a set of 32 rectangles (4 colours and 8 numbers). This process was repeated to measure intelligibility and response time across a range of modulation depths. Additionally, after every 30 trials, the participants were asked to give ratings of effort on a self-report questionnaire known as NASA task load index and these data were also compared across ANC conditions.
- The results showed a small improvement in intelligibility and small deterioration (increase) in response time when the ANC was on. Overall, these findings suggested some complex effects being caused by the ANC in the chimera perception by human listeners.

High Resolution Source Localization using Non Uniform Arrays

Location: Audio Signal Processing Lab, IIT Gandhinagar

Jan. 2021 – Apr. 2021

- Developed a high-resolution source localization algorithm using the concept of the group delay of the coarray MUSIC spectrum.
- Derived a theoretical proof that proved the high-resolution property of the proposed algorithm.

Efficient DOA Estimation using Faster Coarray Interpolation Technique

Location: Audio Signal Processing Lab, IIT Gandhinagar

Jan. 2020 – May. 2020

- Developed a faster coarray interpolation technique using the truncated nuclear norm regularisation technique, which results in an efficient DOA estimation using non-uniform linear arrays.
- Derived the convergence criterion for the proposed algorithm.

RESEARCH INTERESTS

- Multi-channel/Array Signal Processing
- Spherical Microphone Array Processing
- Acoustic Beamforming
- Automatic Speech Recognition
- Hearing Aids and Binaural Beamforming
- Speech Enhancement based on statistical signal processing
- Sound Source Localization and DOA Estimation

PUBLICATIONS

Journal Articles

1. **S. K. Yadav**, S. R. M. Prasanna and N. V. George, "Near-Field Source Localization and Beamforming in the Spherical Sector Harmonics Domain," in *IEEE Journal of Selected Topics in Signal Processing*, vol. 18, no. 4, pp. 546-560, 2024.
2. **S. K. Yadav** and N. V. George, "Joint Dereverberation and Beamforming with Blind Estimation of the Shape Parameter of the Desired Source Prior," in *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 32, pp. 779-793, 2024.
3. **S. K. Yadav** and N. V. George, "Sparse Distortionless Modal Beamforming for Spherical Microphone Arrays," in *IEEE Signal Processing Letters*, vol. 29, pp. 2068-2072, 2022.
4. **S. K. Yadav** and N. V. George, "Underdetermined DOA Estimation Using Arbitrary Planar Arrays via Coarray Manifold Separation," in *IEEE Transactions on Vehicular Technology*, vol. 71, no. 11, pp. 11959-11971, Nov. 2022.
5. **S. K. Yadav** and N. V. George, "Coarray MUSIC-Group Delay: High-Resolution Source Localization Using Non-Uniform Arrays," in *IEEE Transactions on Vehicular Technology*, vol. 70, no. 9, pp. 9597-9601, Sept. 2021.
6. **S. K. Yadav** and N. V. George, "Underdetermined Direction-of-Arrival Estimation Using Sparse Circular Arrays on a Rotating Platform," in *IEEE Signal Processing Letters*, vol. 28, pp. 862-866, 2021.
7. **S. K. Yadav** and N. V. George, "Fast Direction-of-Arrival Estimation via Coarray Interpolation Based on Truncated Nuclear Norm Regularization," in *IEEE Transactions on Circuits and Systems II: Express Briefs*, vol. 68, no. 4, pp. 1522-1526, April 2021.
8. **S. K. Yadav** and N. V. George, "Enhanced DOA estimation through Coprime Planar Coarray Interpolation via Alternating Projection," accepted for publication in *IEEE Transactions on Vehicular Technology* [In Press].
9. S. Vadhvana, **S. K. Yadav**, S. S. Bhattacharjee and N. V. George, "An Improved Constrained LMS Algorithm for Fast Adaptive Beamforming Based on a Low Rank Approximation," in *IEEE Transactions on Circuits and Systems II: Express Briefs*, vol. 69, no. 8, pp. 3605-3609, Aug. 2022.
10. S. Joel, **S. K. Yadav** and N. V. George, "Coarray LMS: Adaptive Underdetermined DOA Estimation with Increased Degrees of Freedom," in *IEEE Signal Processing Letters*, vol. 31, pp. 591-595, 2024.
11. S. Joel, **S. K. Yadav** and N. V. George, "Adaptive Low-Rank DOA Estimation using Complex Kronecker Product Decomposition," in *IEEE Transactions on Vehicular Technology*, vol. 73, no. 7, pp. 10726-10731, 2024.
12. S. Joel, **S. K. Yadav** and N. V. George, "Enhanced Bias-Compensated NLMS for Adaptive DOA Estimation," in *IEEE Sensors Letters*, vol. 8, no. 5, pp. 1-4, May 2024.
13. S. Joel, M. L. N. S. Karthik, **S. K. Yadav** and N. V. George, "Adaptive Direction of Arrival Estimation Using Sparsity Constrained Complex NLMS Algorithm With Variable Penalty Factor," in *IEEE Transactions on Vehicular Technology*, 2025.
14. S. Joel, **S. K. Yadav** and N. V. George, "Adaptive Root-Finding Framework for DOA Estimation in Uniform Circular Arrays," in *IEEE Transactions on Vehicular Technology*, 2026 [in press]. doi: 10.1109/TVT.2026.3711488
15. S. Joel, **S. K. Yadav**, M. L. N. S. Karthik and N. V. George, "Fourier Domain Gradient Descent Total Least Square/Fourth Algorithm for Efficient Adaptive Direction of Arrival Estimation," in *IEEE Transactions on Vehicular Technology*, vol. 75, no. 2, pp. 2426-2438, 2026.
16. S. Joel, **S. K. Yadav** and N. V. George, "Integrated Approach for Efficient Adaptive Underdetermined DOA Estimation: Coarray LMS With Covariance Matrix Error Removal," in *IEEE Transactions on Vehicular Technology*, vol. 75, no. 1, pp. 1649-1654, 2026.

Conference Proceedings

1. **S. K. Yadav** and N. V. George, "Radial Acoustic Beamforming via Near-Field Array Processing in the Spherical Harmonics Domain," accepted at the *23rd IEEE Statistical Signal Processing Workshop (IEEE SSP)*, 2025.
2. **S. K. Yadav** and N. V. George, "Joint Sparse Support Recovery for Direction of Arrival Estimation Using Rational Arrays," accepted at the *101st IEEE Vehicular Technology Conference (VTC2025-Spring)*, 2025.
3. **S. K. Yadav** and N. V. George, "Third-order Tensor Decomposition based Multichannel Linear Prediction for Robust Dereverberation," in *Proc. of the 18th IEEE International Workshop on Acoustic Signal Enhancement (IEEE IWAENC)*, Aalborg, Denmark, Sep 9-12 2024, pp. 230-234.
4. **S. K. Yadav** and N. V. George, "Speech Enhancement via Maximum Likelihood Modal Beamforming with Complex Gaussian and Laplacian Priors," in *Proc. of the 31st European Signal Processing Conference (EUSIPCO)*, 4 - 8 Sep 2023, Helsinki, Finland, pp. 26-30.
5. **S. K. Yadav** and N. V. George, "Coarray Manifold Separation in the Spherical Harmonics Domain for Enhanced Source Localization," in *Proc. of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 22 - 27 May 2022, Singapore, pp. 5038-5042.

6. **S. K. Yadav** and N. V. George, "Sparse 3D Array Made from Nested Linear Array Branches for Underdetermined Source Localization," in *Proc. of the 30th European Signal Processing Conference (EUSIPCO)*, 29 Aug - 2 Sep 2022, Belgrade, pp. 1746-1750.
7. **S. K. Yadav** and N. V. George, "Distortionless Acoustic Beamforming with Enhanced Sparsity based on Reweighted ℓ_1 -Norm Minimization," in *Proc. of the International Congress on Acoustics (ICA)*, 24 - 28 Oct 2022, South Korea, pp. 227-234.
8. **S. K. Yadav** and B. U. Seeber, "Robustness of Binaural Beamformers to Spatial Steering Estimation Error in Dynamic Situations," in *Proc. of the 52nd Fortschritte der Akustik-DAGA*, Dresden, Germany, 2026, pp. 491-493. DOI: <https://doi.org/10.71568/daga2026.395>.
9. **S. K. Yadav** and J. N. Patel, "Application of Compressed Sensing using a Reed Solomon (RS) code based Deterministic Measurement Matrix," in *Proc. of the 10th International Conference on Computing, Communication and Networking Technologies (ICCCNT)*, IIT Kanpur, India, 2019, pp. 61-65.
10. S. Joel, **S. K. Yadav** and N. V. George, "Closed-Form Adaptive DOA Estimation using Sensor Array over Spherical Sector," in *Proc. of the 29th International Conference on Information Fusion (FUSION 2026)*, Trondheim, Norway [in press].
11. S. Joel, **S. K. Yadav** and N. V. George, "Adaptive Closed-form DOA Estimation in the Spherical Harmonics Domain," in *Proc. of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, Barcelona, Spain, 2026, pp. 20816-20820.
12. J. Jayan, K. Rathod, **S. K. Yadav** and N. V. George, "Acoustic Beamforming using Maximum Likelihood Criteria with Laplacian Model of the Desired Signal," in *Proc. of the 53rd International Congress and Exposition on Noise Control Engineering (INTERNOISE)*, Nantes, France, 25-29 Aug 2024, vol. 270, no. 2, pp. 9479-9489.

TEACHING EXPERIENCE

Teaching Assistantship at TU Munich, Germany

- EI73871 – Technical Acoustics and Noise Abatement, *Instructor: Prof. Dr. -Ing. Bernard U. Seeber*
- Winter Semester 2024-25 and 2025-26

Graduate Teaching Fellow at IIT Gandhinagar, Gandhinagar

- EE 609 – Advanced Signal Processing, *Co-Instructor: Dr. Nithin. V. George*
- Semester II of 2022-23

Teaching Assistantship at IIT Gandhinagar, Gandhinagar

- EE 609 – Advanced Signal Processing, *Instructor: Dr. Nithin. V. George*
- Semester II of 2020-2021 and Semester II of 2021-2022
- EE 411 – Digital Signal Processing, *Instructor: Dr. Nithin. V. George*
- Semester I of 2019-20
- Semester I of 2022-23
- ES 615 – Nature Inspired Computing, *Instructor: Dr. Nithin. V. George*
- Semester II of 2019-20
- Semester I of 2020-2021
- Semester I of 2021-2022

Teaching Assistantship at SVNIT, Surat

- EC 402 – Optical Communication Lab, *Instructor: Dr. Piyush N. Patel*
- Semester I of 2017-18
- VL 311 – Digital Electronics Lab, *Instructor: Dr. Anand D. Darji*
- Semester II of 2017-2018

PRE-PHD WORK EXPERIENCE

Idea Cellular Limited, Pune, India

Assistant Manager, Intelligent Networks Division

Jul. 2015 – May. 2016

- Worked in the Intelligent Network (IN) division of the Network Services Department of Idea Cellular Ltd., Pune, India.
- I was part of the operations team of the department that involved maintaining connectivity between various IN nodes such as CCN, SDP, MINSAT etc. and also between the main fibre optical network and the IN nodes.

Lantiq Communication Pvt. Ltd., Bangalore, India

Undergraduate Intern, Chip Verification Team

Jul. 2014 – Dec. 2014

- Lantiq makes semiconductor products for computer networks used by common carrier telecommunications companies in access networks and home networking. It was acquired by Intel Corporation in 2015.

- Worked on a project titled 'Formal Verification - Property and Equivalence checking for Wave500'. The work included writing test cases for the verification of various components of the chip and also to write and develop functional coverage models. I also performed Logic Equivalence checks (LEC), Clock domain crossing (CDC) checks, Formal connectivity checks, Lint checks on an active project of the company.

Honda Motors, Gurgaon, India

Undergraduate Intern, Assembly Line Division

May. 2013 – Jul. 2013

- Worked on a project titled 'Study of Electrical Breakdowns in CNC and PLC Machines'.

TECHNICAL SKILLS

- **Programming Languages:** Proficient: MATLAB, Python Basic: VHDL, C, C++
- Proficient in handling Speedgoat real-time target machines for testing real-time performance of signal processing algorithms.

HONOURS AND AWARDS

- Recipient of the PhD Research Fellowship from the Ministry of Human Resource Development, Government of India (2019-2023).
- Recipient of the SPS Travel Grant from the IEEE Signal Processing Society (SPS) to attend and present at the ICASSP conference held in Singapore from May 23-27, 2022.
- Recipient of the SERB International Travel Scheme to attend and present at the EUSIPCO conference held in Belgrade, Serbia, from 29 Aug - 2 Sep 2022.
- Best poster award at the IIT Gandhinagar Post Graduate Research Showcase event in January 2023.
- Recipient of the Postgraduate Research Fellowship from the Ministry of Human Resource Development, Government of India (2017-2019).
- Appeared in the Overall Dean's List at the SVNIT convocation for excellence in Master's studies in 2019.
- Secured First position (rank 1) in order of merit among the successful candidates in the entire state of Assam in the Higher Secondary Examination (2011) conducted by the AHSEC education board, Assam, India.